

PADHAISHURU

GATE Preparation Platform

GATE XE

Engineering Sciences

TREND-BASED MOCK PAPER

— 04 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 20 questions (31%)

Moderate: 32 questions (49%)

Difficult: 13 questions (20%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE XE — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

1M [no -ve]

[Basic Electrical]

V =:

*Enter your answer here***Q2.**

1M [-0.33]

*[General Aptitude]*If $x^2=16$, $x =$:

- (A) 4
- (B) -4
- (C) ± 4
- (D) 0

Q3.

1M [-0.33]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q4.

1M [-0.33]

*[General Aptitude]*If $x^2=16$, $x =$:

- (A) 4
- (B) -4
- (C) ± 4
- (D) 0

Q5.

1M [-0.33]

[Environmental Science]

PM2.5 particle size:

- (A) $<2.5; \mu\text{m}$
- (B) $<5; \mu\text{m}$
- (C) $<10; \mu\text{m}$
- (D) $<1; \mu\text{m}$

Q6.

1M [-0.33]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q7.

1M [-0.33]

[Fluid Mechanics]

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

Q8.

1M [no -ve]

[Basic Electrical]

V =:

Enter your answer here

Q9.

1M [no -ve]

[Environmental Science]

PM2.5 particle size:

Enter your answer here

Q10.

1M [no -ve]

[Thermodynamics]

First law:

Enter your answer here

Q11.

1M [-0.33]

*[General Aptitude]*If $x^2=16$, $x =$:

- (A) 4
- (B) -4
- (C) ± 4
- (D) 0

Q12.

1M [-0.33]

*[General Aptitude]*If $x^2=16$, $x =$:

- (A) 4
- (B) -4
- (C) ± 4
- (D) 0

Q13.

1M [-0.33]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q14.

1M [-0.33]

[Engineering Mathematics]

If A is 2x2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

Q15.

1M [no -ve]

[Environmental Science]

PM2.5 particle size:

*Enter your answer here***Q16.**

1M [no -ve]

[Basic Electronics]

Diode used as:

*Enter your answer here***Q17.**

1M [no -ve]

[Solid Mechanics]

Tensile stress formula:

*Enter your answer here***Q18.**

1M [-0.33]

*[General Aptitude]*If $x^2=16$, $x =$:

- (A) 4
- (B) -4
- (C) ± 4
- (D) 0

Q19.

1M [no -ve]

[Fluid Mechanics]

Bernoulli assumes:

*Enter your answer here***Q20.**

1M [-0.33]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q21.

1M [-0.33]

[Environmental Science]

PM2.5 particle size:

- (A) $<2.5\mu\text{m}$
- (B) $<5\mu\text{m}$
- (C) $<10\mu\text{m}$
- (D) $<1\mu\text{m}$

Q22.

1M [-0.33]

[Solid Mechanics]

Tensile stress formula:

- (A) F/A
- (B) $F \cdot A$
- (C) F/A^2
- (D) A/F

Q23.

1M [no -ve]

[Computers and Programming]

Binary search complexity:

Enter your answer here

Q24.

1M [-0.33]

[Environmental Science]

PM2.5 particle size:

- (A) $<2.5\mu\text{m}$
- (B) $<5\mu\text{m}$
- (C) $<10\mu\text{m}$
- (D) $<1\mu\text{m}$

Q25.

1M [no -ve]

*[General Aptitude]*If $x^2=16$, $x =$:

Enter your answer here

Q26.

1M [no -ve]

[Material Science]

Thermoplastics:

Enter your answer here

Q27.

1M [-0.33]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q28.

1M [-0.33]

*[Engineering Mathematics]*If A is 2×2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

Q29.

1M [-0.33]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q30.

1M [no -ve]

[Material Science]

Thermoplastics:

*Enter your answer here***Q31.**

2M [no -ve]

[Basic Electrical]

V =:

- (A) IR
- (B) I/R
- (C) R/I
- (D) I^2R

Q32.

2M [-0.66]

[Computers and Programming]

Binary search complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(1)$

Q33.

2M [no -ve]

[Computers and Programming]

Binary search complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(1)$

Q34.

2M [-0.66]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q35.

2M [no -ve]

[Basic Electrical]

V =:

- (A) IR
- (B) I/R
- (C) R/I
- (D) I^2R

Q36.

2M [no -ve]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q37.

2M [-0.66]

[Computers and Programming]

Binary search complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(1)$

Q38.

2M [-0.66]

[Computers and Programming]

Binary search complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(1)$

Q39.

2M [no -ve]

[Fluid Mechanics]

Bernoulli assumes:

Enter your answer here

Q40.

2M [-0.66]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q41.

2M [-0.66]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q42.

2M [no -ve]

[Basic Electronics]

Diode used as:

*Enter your answer here***Q43.**

2M [-0.66]

[Material Science]

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

Q44.

2M [no -ve]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q45.

2M [no -ve]

[Computers and Programming]

Binary search complexity:

*Enter your answer here***Q46.**

2M [-0.66]

[Fluid Mechanics]

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

Q47.

2M [no -ve]

*[General Aptitude]*If $x^2=16$, $x =$:

Enter your answer here

Q48.

2M [-0.66]

*[General Aptitude]*If $x^2=16$, $x =$:

- (A) 4
- (B) -4
- (C) ± 4
- (D) 0

Q49.

2M [-0.66]

[Engineering Mathematics]

If A is 2x2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

Q50.

2M [-0.66]

[Environmental Science]

PM2.5 particle size:

- (A) $<2.5; \mu\text{m}$
- (B) $<5; \mu\text{m}$
- (C) $<10; \mu\text{m}$
- (D) $<1; \mu\text{m}$

Q51.

2M [-0.66]

[Computers and Programming]

Binary search complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(1)$

Q52.

2M [no -ve]

[Environmental Science]

PM2.5 particle size:

- (A) $<2.5; \mu\text{m}$
- (B) $<5; \mu\text{m}$
- (C) $<10; \mu\text{m}$
- (D) $<1; \mu\text{m}$

Q53.

2M [no -ve]

[Environmental Science]

PM2.5 particle size:

- (A) $<2.5\mu\text{m}$
- (B) $<5\mu\text{m}$
- (C) $<10\mu\text{m}$
- (D) $<1\mu\text{m}$

Q54.

2M [-0.66]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q55.

2M [no -ve]

[Engineering Mathematics]

If A is 2x2 with trace 5 and det 6, eigenvalues:

*Enter your answer here***Q56.**

2M [-0.66]

[Engineering Mathematics]

If A is 2x2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

Q57.

2M [-0.66]

*[General Aptitude]*If $x^2=16$, $x =$:

- (A) 4
- (B) -4
- (C) ± 4
- (D) 0

Q58.

2M [-0.66]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q59.

2M [-0.66]

[Solid Mechanics]

Tensile stress formula:

- (A) F/A
- (B) $F \cdot A$
- (C) F/A^2
- (D) A/F

Q60.

2M [-0.66]

*[Engineering Mathematics]*If A is 2×2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

Q61.

2M [no -ve]

[Fluid Mechanics]

Bernoulli assumes:

*Enter your answer here***Q62.**

2M [no -ve]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q63.

2M [no -ve]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q64.

2M [-0.66]

[Material Science]

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

Q65.

2M [-0.66]

[Material Science]

Thermoplastics:

- (A) Thermoset
 - (B) Remoldable
 - (C) Crosslinked
 - (D) None
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